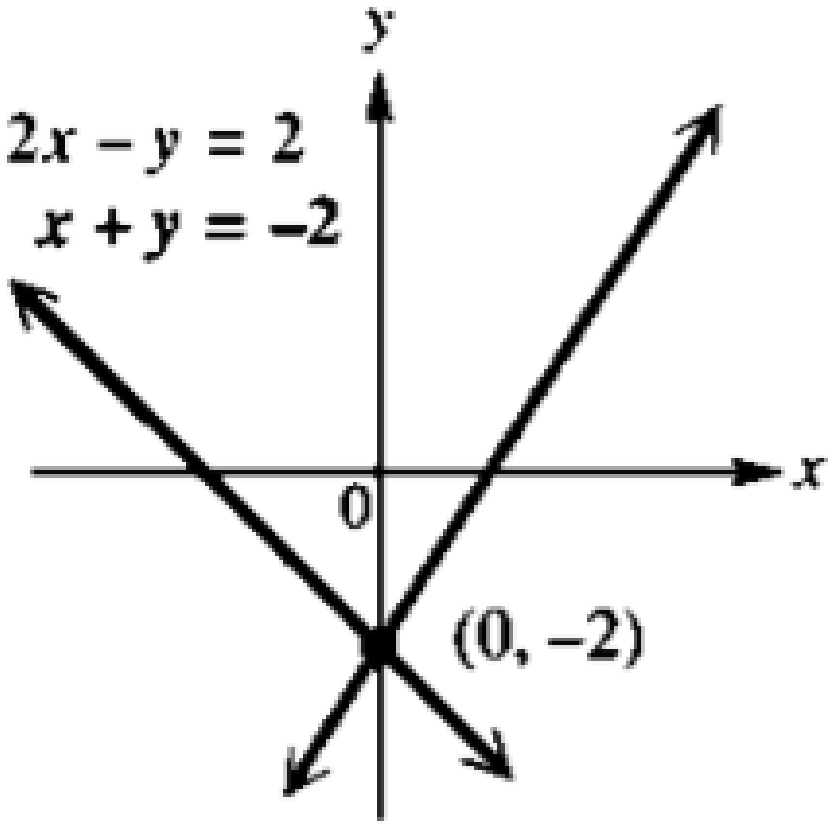
Graphical Method



Step 1

% Define x values

x = -5:0.1:5; %

Step 2

Define the two equations

y1 = 2*x - 2; % 2x - y = 2

y2 = -x - 2; % x + y = -2

Step 3 % Plot the lines

```
figure;
plot(x, y1, 'b-', 'LineWidth', 2);
hold on;
plot(x, y2, 'r--', 'LineWidth', 2);
```

Step 4 % Find and plot the intersection point

```
A = [2 -1; 1 1];
B = [2; -2];
P = A\B;                    % Solution [x; y]
```

Step 5

```
plot(P(1), P(2), 'ko', 'MarkerSize', 8, 'MarkerFaceColor', 'k’);
text(P(1)+0.2, P(2), sprintf('(%.1f, %.1f)', P(1), P(2)));
```

Step 6

```
% Formatting
xlabel('x’);
ylabel('y’);
title('Graphs of 2x - y = 2 and x + y = -2’);
legend('2x - y = 2', 'x + y = -2', 'Intersection', 'Location', 'best’);
grid on;axis equal;
xlim([-5 5]);
ylim([-8 8]);
```

Solving Basic Algebraic Equations

- The **solve** function is used for solving algebraic equations.

S = solve(eqn,var)

- solves the equation **eqn** for the variable var. If you do not specify var, it
- determines the variable to solve for. Before solve the equation, the **syms**
- function is used to define symbolic variables.

Solving System of Equations

The solve function can also be used to generate solutions of systems of equations involving more than one variables.

Example:

$$2x + 3y = 7$$

$$4x - 2y = 2$$

```
syms x y
eqns = [2*x + 3*y == 7, 4*x - 2*y == 2];
sol = solve(eqns, x, y);
disp(sol);
```

Example:

$$2x + 3y - z = 5$$

$$x - y + 2z = 3$$

$$3x + 2y + 2z = 2$$

```
syms x y z
eqns = [2*x + 3*y - z == 5, x - y + 2*z == 3, 3*x + 2*y + 2*z == 2];
sol = solve(eqns, x, y, z);
disp(sol);
```